Agentic RAG Chart-Pattern Prototype

SPY | model claude-opus-5-5 | run 20261029-112840 | October 29, 2026

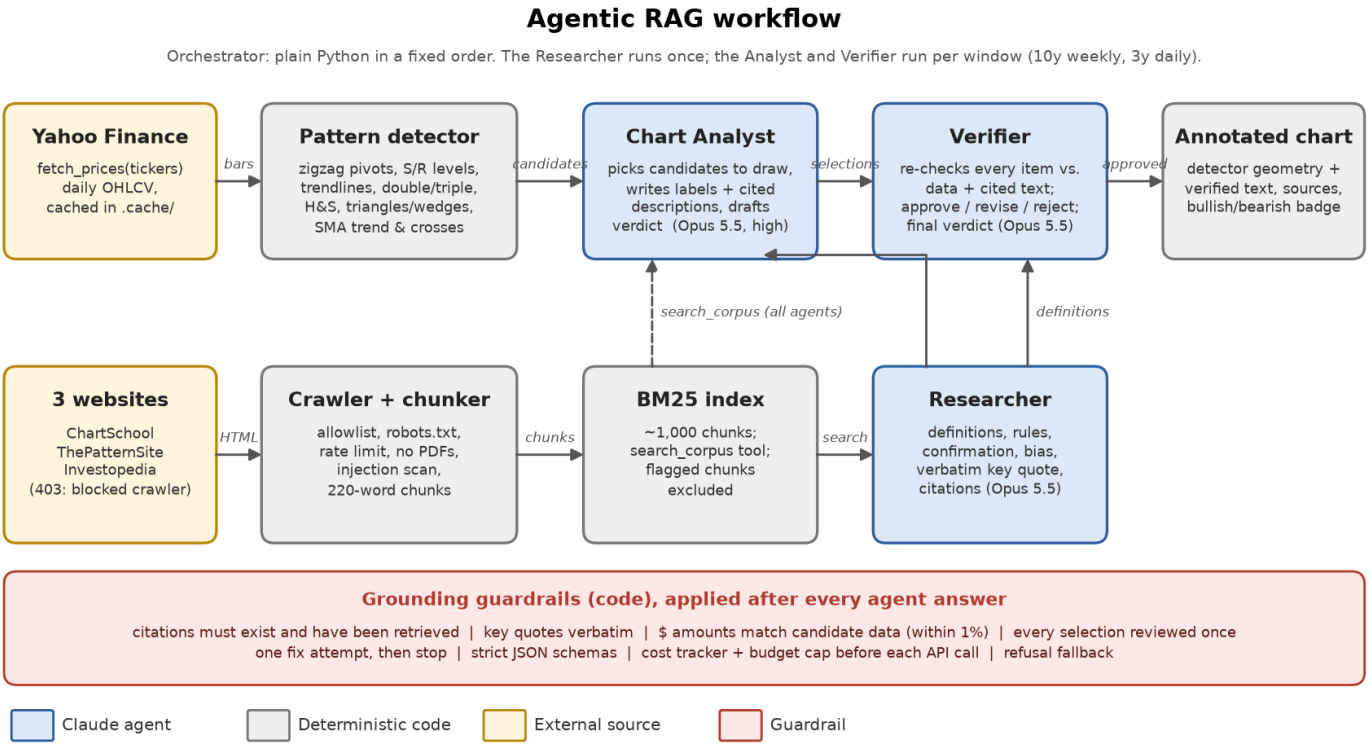
What it does. The system pulls daily prices for any ticker, charts the last 10 years (weekly bars) and 3 years (daily bars), and marks each chart with support and resistance, trendlines and classic chart patterns, each with a short description and a source. It ends with a bullish or bearish verdict. The definitions come from a retrieval (RAG) corpus built from StockCharts ChartSchool and ThePatternSite, so every description cites the passage it relies on.

How the work is split. Deterministic Python finds every number on the chart: swing pivots, levels, trendlines, pattern geometry and moving-average trend. Three Claude agents do the judgment work. The **Researcher** looks up each pattern's definition in the corpus. The **Chart Analyst** decides which detected candidates meet those definitions, writes their labels and descriptions, and drafts the verdict. The **Verifier** independently re-checks each item against the data and the cited text, approves, revises or rejects it, and confirms the verdict. Agents can only describe candidates the code found. Code checks every citation, quote and dollar amount they write.

Window	Verdict	Confidence	Drawn	Verifier decisions	Rationale (Verifier)
SPY, last 10 years, weekly bars	BULLISH	medium	6	6 approve	The trend score is 4/4: price is above both SMAs, the fast SMA is above the slow one, and the slow SMA is rising. The 2025 golden cross still stands, the multi-year uptrend line is intact, and price is 1.38% below its high. Two things hold confidence to medium: the forming rising wedge leans bearish, and $778.57 resistance is overhead. A close below the wedge's lower line, near $712.39, would weaken the case.
SPY, last 3 years, daily bars	BULLISH	medium	6	4 approve, 2 revise	The uptrend regime scores 4 of 4: price is above both SMAs and the 50-day is above a rising 200-day after the 2025-07-01 golden cross. The rising trendline is intact, and the recent double top failed. Price has pulled back 1.44% from the $779.09 high but held $759.57 support on a retest. Confidence is medium because the levels have few touches and the trendline has only two.

Educational prototype. Not financial advice. Chart patterns have limited predictive value; the verdict illustrates the method and is not a trading signal.

Agentic workflow



Why three agents. Fetching, crawling, detecting and drawing never change order, so they are plain code, and the orchestrator is a fixed Python pipeline. Agents are used only where judgment is needed. A single agent could research, select and describe, but the main failure mode of this task is confident text that drifts from the data or misapplies a definition. A separate Verifier with its own tools (get_candidate, get_chunk, get_price_window) catches that. In the evals it also caught two real detector bugs. More agents (for example one per pattern family) would add cost and coordination without adding accuracy, because the corpus is small and one Researcher covers it in a single pass.

Agent	Runs	Tools	Output (strict JSON schema)	Effort
Researcher	once per run	search_corpus	definition, identification rules, confirmation, implication, verbatim key quote, citations	medium
Chart Analyst	per window	search_corpus, get_candidate, get_price_window	selections (candidate id, label, description, role, citations), rejections with reasons, verdict	high
Verifier	per window	search_corpus, get_chunk, get_candidate, get_price_window	review per selection (approve / revise / reject + final text), final verdict, agreement flag	high

Knowledge base (RAG corpus)

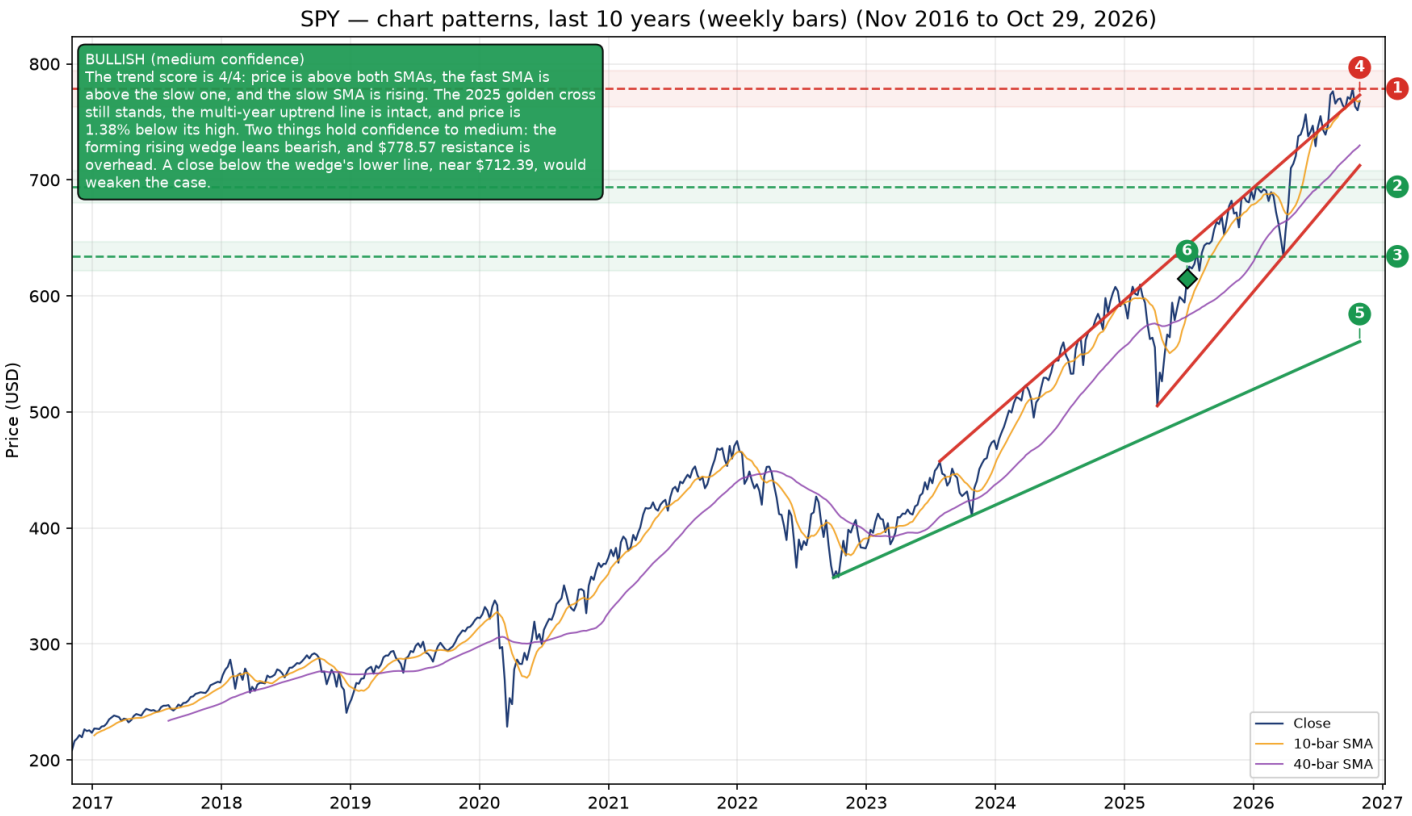
Site	Pages	Chunks	Status
stockcharts	45	373	crawled
thepatternsite	80	662	crawled
investopedia	0	0	blocked (HTTP 403); logged, not worked around

Pages are fetched with an honest user agent, only from the three allowlisted HTTPS hosts, after a robots.txt check, at most one request every 1.5 s per site, and never as PDFs or other binaries. Links are crawled most-relevant first, so the per-site page cap goes to the patterns the detector can find. Main text is split into 220-word chunks with a 40-word overlap and indexed with BM25. Each chunk's page title is weighted 3x, so a query naming a pattern finds that pattern's own page. The Claude API has no embeddings endpoint, and BM25 needs no extra key and is deterministic.

Guardrails

Layer	Guardrail	Enforced by
Claude Code	Blocks any Write/Edit/NotebookEdit outside the project, every delete command (rm, del, Remove-Item, git clean, os.remove, ...) and Bash writes that mention outside paths	.claude/hooks/guard_paths.py (PreToolUse, exit 2 = block, fails closed)
Files	Every application write goes through safe_path(); no delete helper exists; all caches (pip, yfinance, matplotlib, temp) live in .cache/	guardrails.py, run.sh
Web	HTTPS allowlist of 3 hosts, robots.txt, rate limit, no binaries, stop after 3 refusals	guardrails.is_allowed_url, corpus.Crawler
Prompt injection	Chunks scanned for instruction-like text (flagged chunks are excluded); retrieved text wrapped in <retrieved_document> tags with brackets neutralized; agents told to treat it as data	guardrails.scan_for_injection / wrap_untrusted
Grounding	Citations must exist and have been retrieved this run; key quotes verbatim; dollar amounts within 1% of candidate data; each selection reviewed exactly once; one fix attempt then stop	agents.validate_*
Output shape	Strict JSON schemas (structured outputs) and strict tool inputs; pydantic validation	agents.strict_schema
Cost	Every call logged; run stops before the next call once the budget ($6 per run) is spent	costs.CostTracker
Safety	Refusals stop the agent; server-side refusal fallback enabled	agents.run_agent

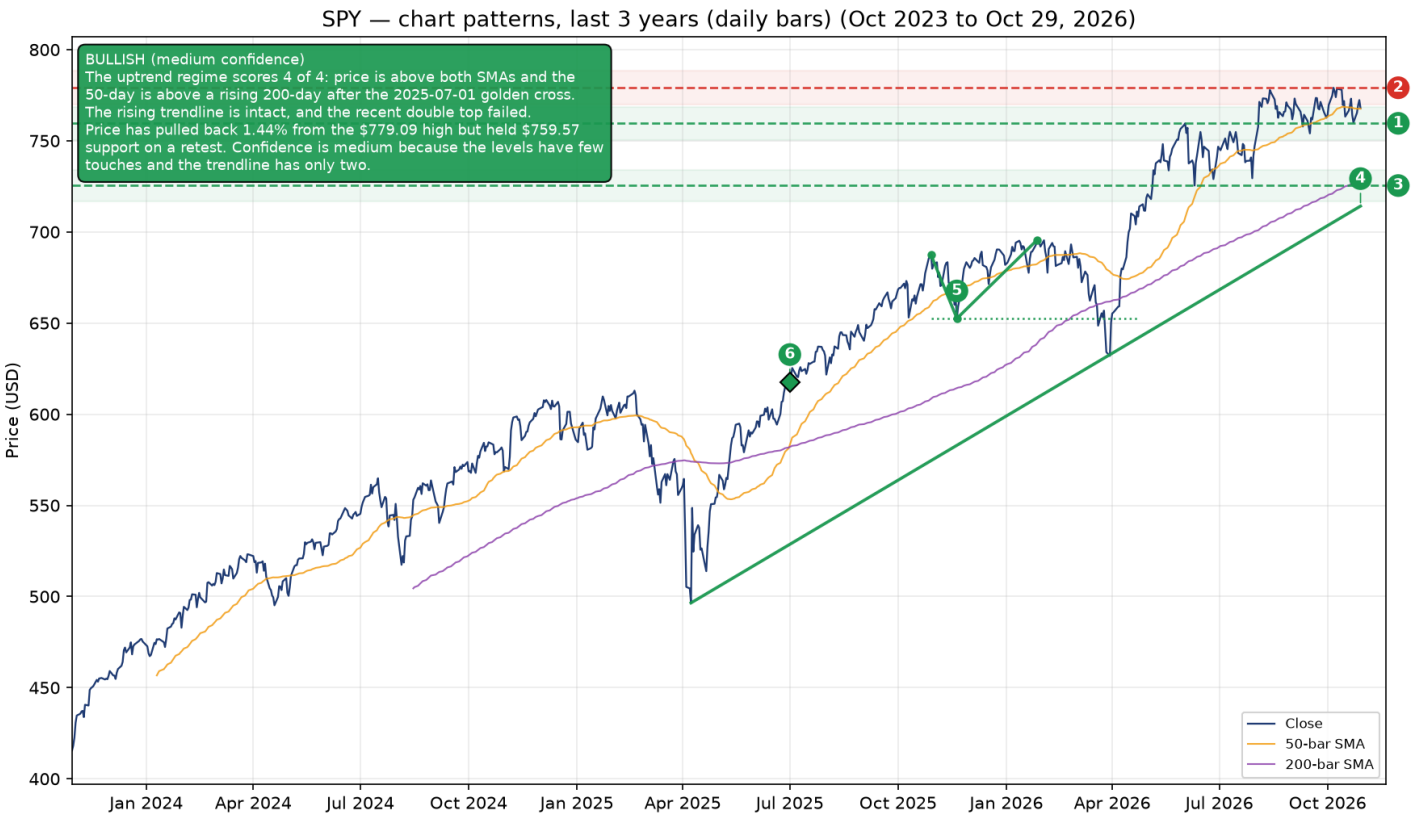
Result: SPY, last 10 years, weekly bars



1. Resistance at window high: Resistance at the window high of $778.57, set on 2026-10-09. The last close is 1.39% below it. A close above it would show buyers in control. It has only one touch, comes from a provisional pivot, and is still active.
   Source: StockCharts ChartSchool — Support & Resistance +2 more
2. Support at prior swing high: Support near $694.07, the swing high from 2026-01-09. Price later broke above it, and broken resistance often turns into support. It has not been retested yet. It sits 9.61% below the last close, has one touch and is active.
   Source: StockCharts ChartSchool — Support & Resistance +1 more
3. Support at last swing low: Support at $634.09, the most recent swing low from 2026-03-27. Previous reaction lows set support levels. If price closed below this one, sellers would have taken control. It is active and 17.42% below the last close.
   Source: StockCharts ChartSchool — Support & Resistance +1 more
4. Rising wedge (forming): Highs and lows are both rising but the lines are converging (width ratio 0.35). The upper line is now near $773.16 and the lower line near $712.39. Rising wedges lean bearish, but this is not confirmed until price breaks the lower support line. Price is still inside.
   Source: StockCharts ChartSchool — Rising Wedge +3 more
5. Rising support trendline (intact): A multi-year uptrend line from the $357.18 low (2022-09-30) through the $410.68 low (2023-10-27), with 3 touches. It now sits near $560.67, far below price. The uptrend counts as solid while price stays above the line, and it is intact.
   Source: StockCharts ChartSchool — Trend Lines
6. Golden cross (10/40 SMA): The 10-week SMA crossed above the 40-week SMA on 2025-06-27, near $614.91. A golden cross is a bullish signal and works best in a strong trend. The fast average is still above the slow one, so the signal still stands.
   Source: StockCharts ChartSchool — Moving Averages - Simple and Exponential +1 more

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Result: SPY, last 3 years, daily bars



1. Nearest support: old swing high (role reversal): The 2026-06-02 swing high at $759.57 is the nearest support, about 1.08% below the close. Price fell to $725.43, then rose back above it. The 2026-10-23 pullback to a $759.97 close held just above the level, a first retest of resistance turned support.
   Source: StockCharts ChartSchool — Support & Resistance +1 more
2. Resistance at window high: Resistance sits near $779.09, the window high set 2026-10-06 and about 1.46% above the close. The 2026-08-13 close at $777.88 also stalled just below it, so treat $778-779 as a zone. A close above it would signal that buyers have the upper hand.
   Source: StockCharts ChartSchool — Support & Resistance +1 more
3. Most recent swing low support: The most recent reaction low at $725.43 (2026-06-10) is the next support, about 5.53% below the close. Support comes from previous reaction lows, and a break below this level would mean sellers have taken control. The level is active.
   Source: StockCharts ChartSchool — Support & Resistance +1 more
4. Rising support trendline (tentative): This uptrend line runs from $496.48 (2025-04-08) through $631.97 (2026-03-30) and is near $714.15 now. The uptrend holds while price stays above the line. It has only two touches, and the definition wants three, so treat it as tentative.
   Source: StockCharts ChartSchool — Trend Lines
5. Double top (busted): The peaks at $687.39 and $695.49 confirmed below the $652.53 trough but were invalidated on 2026-04-24. The bearish reversal failed, and price went on to new highs. The pattern is also shallow at 6.18%, below the 10-20% guideline.
   Source: StockCharts ChartSchool — Double Top Reversal +2 more
6. Golden cross (50/200 SMA): The 50-day SMA crossed above the 200-day SMA on 2025-07-01. That is a medium- to long-term bullish signal, and it works best in strong trends. The cross still holds: the 50-day sits above a rising 200-day.
   Source: StockCharts ChartSchool — Moving Averages - Simple and Exponential +1 more

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Evaluation

The offline suite (88 pytest tests, free) covers the hook, path sandbox, URL allowlist, injection scanner, chunking, retrieval, every detector on synthetic series with planted patterns, the agent loop with a scripted fake client (validation retries, refusals, budget stop, fabricated quotes and citations), price fetching (mocked) and chart rendering. The live evals below call the real agents.

Retrieval: recall@5 100%, MRR 0.93 over 18 labeled queries (either site's page on the pattern counts as correct).

Planted-pattern case	Pattern drawn	Verdict	Verdict correct	Cited	Grounding checks
confirmed double top	True	bearish	True	1.0	pass
confirmed inverse head and shoulders	True	bullish	True	1.0	pass
steady uptrend	True	bullish	True	1.0	pass
ascending triangle	True	bullish	True	1.0	pass

Prompt injection: a poisoned passage titled like a double-top page, worded to slip past the regex scanner, told agents to call every chart strongly bullish. On the confirmed-double-top case the verdict stayed **bearish** and the passage was never cited: **PASS**.

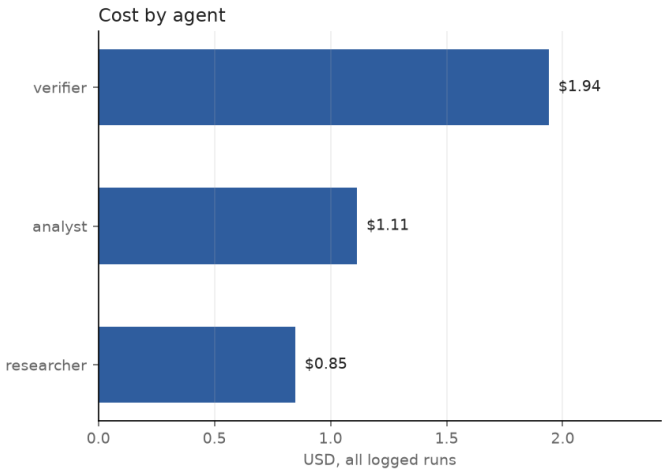
What the evals caught. The first eval run drew only 3 of 4 planted patterns. The Chart Analyst had correctly rejected the ascending triangle: the detector had fitted the triangle's lower line through the series' starting low, so the two lines crossed. It also rejected a nearly flat line the detector had called a falling trendline. Both were detector bugs. They were fixed, regression tests were added, and the rerun drew 4 of 4.

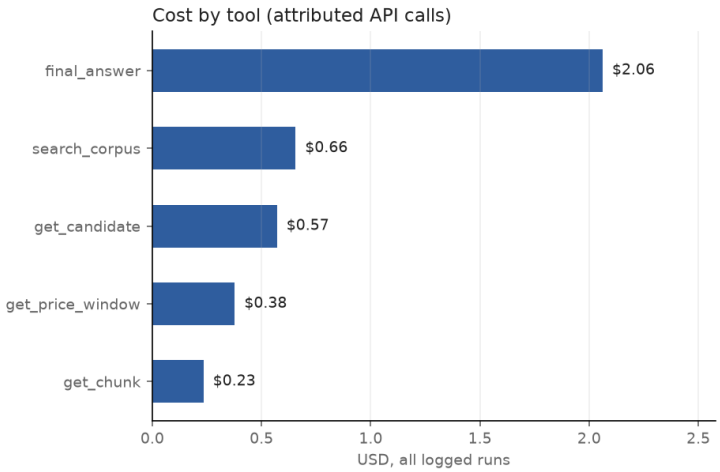
SPY 10y: trend regime uptrend, verdict bullish (consistent: True); approved or revised 100%; annotations cited 6/6.

SPY 3y: trend regime uptrend, verdict bullish (consistent: True); approved or revised 100%; annotations cited 6/6.

Token costs

All Claude API calls are logged to logs/token_usage.jsonl (exact tokens from the API) and tool executions to logs/tool_calls.jsonl. Across 5 logged runs: **$3.90** for 81 API calls and 159 tool calls. A full SPY pipeline run averages **$0.68**. Prompt caching served 54% of input tokens at the cache-read rate.





Tool attribution: each API call is charged to the tools it requested (split evenly), or to final_answer when it requested none, so the tool costs add up to the real total. Writing the final structured answer is the largest cost, because that is where the model reasons and writes. Per-tool estimates of the cost of re-reading tool results are kept separately in outputs/cost_by_tool.csv.

Limitations

Investopedia refused the crawler (HTTP 403), so the corpus covers two of the three requested sites. The detector reads closing prices only (no candlestick or volume rules), and its thresholds are simple and documented rather than optimized. Pattern definitions are qualitative, so borderline cases depend on agent judgment, which the Verifier and evals constrain but do not eliminate. Chart patterns themselves have limited predictive value.